[Insert Project Title Here]

[Insert Full Name Here]

School of [School Name]

College of [College Name]

University of Lincoln

Submitted in partial satisfaction of the requirements for
the Degree of Doctor of Philosophy
in [Subject Area]

Primary Supervisor: [Primary Supervisor Name]
Secondary Supervisor: [Secondary Supervisor Name]
Third Supervisor: [Third Supervisor Name]

[Month Year]

Declaration

I declare that this thesis is my own work and has not been submitted for any other academic award. Where material has been derived from other sources, this has been indicated in the text and in the list of references.

Signed: Date:

[Your Full Name]

Acknowledgements

Write your acknowledgements here. You may wish to thank your supervisors, participants, friends, and family.

Abstract

Write your abstract here. Summarise the research problem, methodology, key findings, and conclusions.

Keywords: [Keyword 1], [Keyword 2], [Keyword 3], [Keyword 4], [Keyword 5]

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List of Acronyms

API – Application Programming Interface

ML – Machine Learning

Chapter 1

Introduction

The introduction to a PhD thesis should set the scene for the entire body of work. It should clearly articulate the problem being addressed, motivate the research, and provide the reader with a roadmap of the thesis. Unlike an MSc dissertation, a PhD introduction is expected to establish the broader significance of the research and its intended contributions to the field.

This document has been generated using Typst, a modern typesetting system. The subsections below demonstrate some basic Typst commands and formatting features, alongside the recommended structure for a PhD introduction.

1.1 Motivation

Clearly define the problem or gap in knowledge that your thesis addresses. This section should motivate the research by explaining why this problem is important and what consequences arise from it remaining unsolved.

1.2 Aim and Objectives

State the overall aim of your research and the specific objectives you set out to achieve. The aim is the broad goal of your research, while the objectives are the specific steps you will take to achieve that aim.

1.3 Research Questions

State your overarching research questions. These should flow naturally from the problem statement and guide the structure of your experimental work. For example:

- **RQ1:** [Your first research question]

- **RQ2:** [Your second research question]

1.4 Thesis Structure

Provide a chapter-by-chapter summary of the thesis structure so the reader understands how the work is organised and how each chapter contributes to answering the research questions.

- **Chapter 2: Related Work** — [Brief description]
- **Chapter 3: [Experimental Chapter Title]** — [Brief description]
- **Chapter 4: [Experimental Chapter Title]** — [Brief description]
- **Chapter 5: [Experimental Chapter Title]** — [Brief description]
- **Chapter 6: Main Findings** — [Brief description]
- **Chapter 7: Conclusions and Future Work** — [Brief description]

1.5 Contributions

Summarise the key contributions of this thesis. This section gives the reader an early overview of what the thesis achieves. Contributions may include new frameworks, empirical findings, software artefacts, datasets, or methodological advances.

List of Publications

If you have published any work during your PhD, list your publications here. This is a common inclusion in PhD theses and demonstrates your contribution to the field. Group publications by type if appropriate.

Journal Articles

[Your Name], [Co-author Name], and [Co-author Name] (Year). Title of journal article. *Journal Name*, Volume(Issue), Pages. DOI.

Conference Papers

[Your Name] and [Co-author Name] (Year). Title of conference paper. In *Proceedings of Conference Name*, Pages. Publisher.

Workshop Papers and Posters

[Your Name] (Year). Title of workshop paper. In *Workshop/Poster Proceedings*, Conference Name.

1.7 Formatting Mathematics

Here are two equations:

$$a = b + 1$$

$$\frac{\hbar^2}{2m}\nabla^2\Psi + V(\boldsymbol{r})\Psi = -i\hbar\frac{\partial\Psi}{\partial t}$$

Inline mathematics can be embedded within body text, for example (x, y), γ, and ρ.

1.8 Figures and Tables

Here is a sentence, and you can see a nice picture in Fig. 1.

Fig.1. An example figure demonstrating image placement.

Also, a table can be found in Table 1.

Table1. An example table demonstrating data presentation.

First Name	Last Name	Age
Bob	Bobbington	24
Benth	Wavies	49
Joe	Bloggs	37
Billy	Bob	10

1.9 Referencing

The included Harvard citation style file means that referencing will follow a Harvard style. You can cite references inline using the `@citekey` syntax. For example, here is a citation (Smith and Doe, 2024). You can also reference conference papers (Williams and Johnson, 2023) and books (Brown, 2022).

This template uses a BibTeX (`.bib`) file for referencing. It is recommended to use a reference manager such as Zotero to export your references as a `.bib` file. The included citation style file follows the Harvard referencing convention. Confirm the required referencing style with your supervisory team, as conventions may vary between disciplines.

Chapter 2

Background

2.1 Introduction

The background chapter (sometimes titled "Literature Review" or "Related Work") is an essential component of any doctoral thesis. A comprehensive review of the literature will provide background to the research, situate the work within the broader field, and identify gaps in the existing knowledge that the thesis aims to address. It should demonstrate a deep and critical engagement with the relevant literature.

At PhD level, this chapter is expected to provide a thorough and critical synthesis of the state of the art. It should not simply summarise existing work, but identify themes, contradictions, and open questions across the literature. The review should clearly establish the theoretical and empirical foundations upon which your experimental chapters build.

It may be helpful to break up this chapter into thematic sections, with each focused on a different area relevant to your research. For example, you might cover the core domain, related methodologies, theoretical frameworks, and adjacent fields.

2.2 [Theme 1]

2.2.1 [Theme 1.1]

[Review and critically discuss a major theme from the literature.]

2.2.2 [Theme 1.2]

[Review and critically discuss a major theme from the literature.]

2.3 [Theme 2]

2.3.1 [Theme 2.1]

[Review and critically discuss another theme, drawing connections to Theme 1 where appropriate.]

2.3.2 [Theme 2.2]

[Review and critically discuss another theme, drawing connections to Theme 1 where appropriate.]

2.4 [Theme 3]

2.4.1 [Theme 3.1]

[Review and critically discuss another theme, drawing connections to Theme 1 where appropriate.]

2.4.2 [Theme 3.2]

[Review and critically discuss another theme, drawing connections to the preceding sections where appropriate.]

Chapter 3

Experimental Chapter 1

Each experimental chapter should be a self-contained piece of work that addresses a specific research question or hypothesis. The structure below is a suggestion; adapt it to suit your research. A PhD thesis may contain multiple studies within a single chapter, or dedicate one chapter per study — discuss the best approach with your supervisor.

3.1 Introduction

Introduce the research question or hypothesis addressed in this chapter, and explain how it relates to the overall thesis aims and the gaps identified in the literature review.

3.2 Related Work

If this chapter draws on literature not covered in the main literature review (e.g., specific to this study's methodology or domain), include a focused related work section here. This keeps each chapter self-contained while avoiding unnecessary repetition of the main review.

3.3 Methodology

Describe the research methodology and experimental design used in this chapter.

3.3.1 Research Questions and Hypotheses

State the specific research questions and/or hypotheses for this study.

3.3.2 Measures and Instruments

Describe the instruments, questionnaires, or metrics used to collect data.

3.3.3 Participants and Procedure

Detail participant recruitment (including sample size justification), demographics, and the step-by-step experimental procedure.

3.3.4 Data Analysis

Describe the analytical methods used (statistical tests, qualitative coding, etc.) and justify their selection.

3.4 Implementation

If this chapter involves a software artefact, system, or tool, detail its design and implementation here. Include architecture overviews, key design decisions, and any relevant technical details. For studies using existing systems, describe the experimental setup and configuration.

3.5 Results

Present the results of the study with supporting figures, tables, and statistical analyses. Organise by research question or hypothesis where appropriate.

3.6 Summary of Findings

Briefly summarise the key findings of this study before moving to discussion. This is particularly helpful in chapters containing multiple sub-studies.

3.7 Discussion

Discuss the implications of the results, how they relate to the literature, and what they mean for the overall thesis. Consider how

these findings address (or fail to address) the research questions posed in this chapter.

3.8 Limitations

Acknowledge the limitations specific to this study. Being transparent about limitations strengthens the academic rigour of the work and helps frame the contributions appropriately.

3.9 Conclusion

Briefly conclude this chapter, summarising its contribution and providing a transition to the next chapter.

Chapter 4

Experimental Chapter 2

This chapter should build upon the findings of the previous experimental chapter. Explain how the research progresses and what new questions or refinements are being addressed.

4.1 Introduction

[Introduce the research question or hypothesis addressed in this chapter, and how it builds on prior work in this thesis.]

4.2 Related Work

[If needed, include focused related work specific to this chapter's study.]

4.3 Methodology

4.3.1 Research Questions and Hypotheses

[State the specific research questions and/or hypotheses for this study.]

4.3.2 Measures and Instruments

[Describe the instruments, questionnaires, or metrics used.]

4.3.3 Participants and Procedure

[Detail participant recruitment, demographics, and the experimental procedure.]

4.3.4 Data Analysis

[Describe the analytical methods used and justify their selection.]

4.4 Implementation

[If applicable, detail the design and implementation of any artefact, system, or experimental setup used in this chapter.]

4.5 Results

[Present the results with supporting figures, tables, and statistical analyses.]

4.6 Summary of Findings

[Briefly summarise the key findings before moving to discussion.]

4.7 Discussion

[Discuss the implications of the results and how they relate to the literature and the overall thesis.]

4.8 Limitations

[Acknowledge the limitations specific to this study.]

4.9 Conclusion

[Briefly conclude this chapter and provide a transition to the next.]

Chapter 5

Experimental Chapter 3

This chapter should represent the final major experimental contribution of the thesis. It may address remaining research questions, extend previous findings to new contexts, or provide a culminating study.

5.1 Introduction

[Introduce the research question or hypothesis addressed in this chapter, and how it completes or extends the experimental narrative of the thesis.]

5.2 Related Work

[If needed, include focused related work specific to this chapter's study.]

5.3 Methodology

5.3.1 Research Questions and Hypotheses

[State the specific research questions and/or hypotheses for this study.]

5.3.2 Measures and Instruments

[Describe the instruments, questionnaires, or metrics used.]

5.3.3 Participants and Procedure

[Detail participant recruitment, demographics, and the experimental procedure.]

5.3.4 Data Analysis

[Describe the analytical methods used and justify their selection.]

5.4 Implementation

[If applicable, detail the design and implementation of any artefact, system, or experimental setup used in this chapter.]

5.5 Results

[Present the results with supporting figures, tables, and statistical analyses.]

5.6 Summary of Findings

[Briefly summarise the key findings before moving to discussion.]

5.7 Discussion

[Discuss the implications of the results and how they relate to the literature and the overall thesis.]

5.8 Limitations

[Acknowledge the limitations specific to this study.]

5.9 Conclusion

[Briefly conclude this chapter and provide a transition to the synthesis chapters that follow.]

Chapter 6

Main Findings

This chapter synthesises the findings from all experimental chapters into a cohesive narrative. Rather than repeating results, it should draw out the overarching themes, patterns, and contributions that emerge when the individual studies are considered together.

6.1 [Topic or Research Question 1]

Synthesise the findings relevant to your first major research question or theme across all experimental chapters. Discuss how the evidence from different studies converges or diverges.

6.2 [Topic or Research Question 2]

Synthesise the findings relevant to your second research question or theme. Draw connections between the individual study results.

6.3 [Additional Topics as Needed]

Continue for each major research question or theme.

6.4 Summary

Provide a concise summary of the main findings of the thesis as a whole, drawing together the threads from the sections above. This should give the reader a clear picture of what the thesis has found before moving to a discussion of wider implications.

Chapter 7

Conclusions and Future Work

This final chapter should bring the thesis to a close. It should not introduce new results, but rather reflect on the work as a whole, its contributions, and where it leads next.

7.1 Summary of Contributions

Summarise the key contributions of this thesis and how they advance the state of the art. Revisit the research questions from the introduction and briefly state how each has been addressed.

7.2 Limitations

Discuss the overarching limitations of the research programme as a whole (individual study limitations should be covered in their respective chapters). Consider constraints on generalisability, methodological limitations, and any threats to validity.

7.3 Future Work

Propose concrete directions for future research that build upon the work presented in this thesis. These should flow naturally from both the findings and the limitations. Consider short-term extensions (e.g., replicating studies, expanding sample sizes) and longer-term research directions.

7.3.1 [Future Direction 1]

[Describe a specific direction for future work.]

7.3.2 [Future Direction 2]

[Describe another direction.]

7.4 Final Remarks

Provide your final concluding remarks. Reflect on the journey of the thesis and its significance for the field.

References

Brown, R. (2022) *An Example Textbook*. Lincoln, UK: University Press

Smith, J. and Doe, J. (2024) An Example Journal Article. *Journal of Example Research,* 42(3) 100–115.

Williams, B. and Johnson, S. (2023) An Example Conference Paper. In: *Proceedings of the International Conference on Examples.* 2023 50–62

Appendices

This section is for any supplementary content that does not fit easily within the main thesis. Examples include questionnaires, extra data, supplementary materials, or detailed proofs.

Appendix A: Code Snippets with Codelst

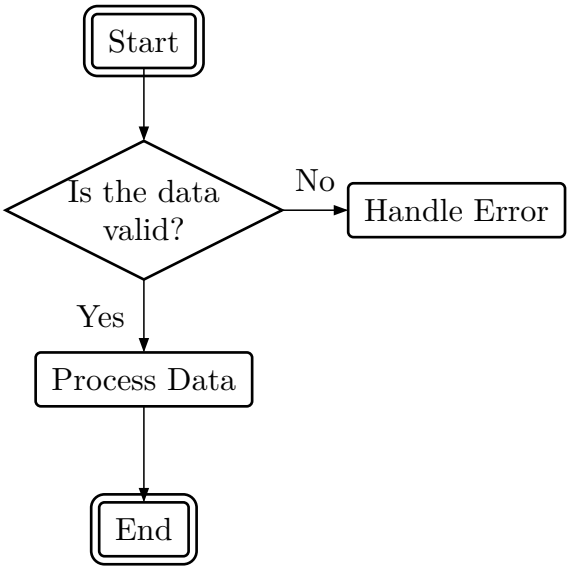
Here you can see a short snippet of code that was used to implement a simple Python sorting algorithm. The Codelst package provides syntax highlighting for code blocks:

```
def bubble_sort(array):
    n = len(array)

    for i in range(n):

        already_sorted = True

        for j in range(n - i - 1):
            if array[j] > array[j + 1]:

                array[j], array[j + 1] = array[j + 1], array[j]

                already_sorted = False

        if already_sorted:
```

```
            break
    return array
```

Appendix B: Flowcharts with Fletcher

This uses the Fletcher package to create a flowchart. The package is available at: https://typst.app/universe/package/fletcher/



Word Count Summary

(excluding figures, tables, captions, and references)

Chapter	**Word Count**
Introduction	530
Background	246
Experimental Chapter 1	364
Experimental Chapter 2	191
Experimental Chapter 3	199
Main Findings	149
Conclusions and Future Work	180
Total	**1859**

Overall Word Count: 1859 words